MANUFACTURING EXECUTION SYSTEM (MES) CONCEPT

Advanced Assembly Line Scheduling Engine

Finite capacity planning, SAP routing synthesis, and multi-line operational resilience.

The Real-World Production Challenge

Why standard theoretical scheduling fails on physical shop floors.

Setup Overhead

Sequence-dependent changeovers create massive downtime between variants.

Capacity Friction

Machine bottlenecks and labor constraints fragment continuous production flow.

Shift Boundaries

Jobs cut across shifts, breaks, and weekends requiring precise WIP tracking.

SAP Routing & Master Data Integration

Translating enterprise structures into actionable shop floor timing.

TABLE PLPO

Operation Routings

Defines sequence, work center, setup duration, and machine run times.

WORK CENTERS

Resource Pools

Maps available machine counts and mandatory operator headcount.

FACTORY CALENDAR

Shift Windows

Establishes active working intervals, breaks, and planned maintenance.

Finite Capacity & Setup Optimization

Greedy dispatching logic balancing completion time and changeover cost.

01

State Lookups

Evaluates current machine type against candidate job requirements.

02

Time Slicing

Advances working minutes strictly inside valid shift operating windows.

03

Score Minimization

Selects jobs minimizing both tardiness and excessive retooling delays.

Multi-Line Load Balancing

Distributing production batches across alternative compatible work centers.

Compatibility Filtering

Validates line capability matrices before candidate assignment.

Earliest Completion

Simulates setup and run times across all lines to find the fastest path.

Shift Boundaries & Flow Dynamics

Managing mid-operation cutoffs and conveyor buffer saturation.

Shift Rollovers

Pausing active job clocks cleanly at shift ends without data corruption.

Upstream Starvation

Handling fast upstream machines stalled by slow downstream bottlenecks.

Conveyor Buffers

Respecting physical WIP limits between automated assembly stages.

Disruptions & Quality Exceptions

Automated responses to breakdowns, stockouts, and inspection spikes.

Unplanned Breakdowns

Locking machine states and triggering instant dynamic rescheduling.

Component Stockouts

Pausing operation clocks during feeder reel exhaustion and re-kitting.

AOI Defect Spikes

Isolating offline rework queues and blocking downstream flow.

Partial Batch Carryover Handling

Automating incomplete lot rollovers to the following shift.

Backlog

Splits active work orders into completed quantities and rescheduled backlogs automatically recalculated for next-day execution.

Closed-Loop MES Telemetry

Event-driven feedback loops bridging shop floor sensors and scheduling logic.

OPC UA Triggers

Real-time machine state signals instantly notify the scheduling engine.

Dynamic Re-sequencing

Instantly recalculates unstarted batch priorities across active factory lines.

Robust Assembly Automation

Combining SAP intelligence with finite capacity scheduling transforms shop floor chaos into predictable, optimized throughput.